Gloss-free Sign Language Translation: Improving from Visual-Language Pretraining Introduction

Sign language is the primary communication method used by deaf communities. Sign Language Translation (SLT) aims to translate sign language videos into natural spoken language sentences. However, this task is difficult due to the complex visual-language relationship and the limited availability of annotated datasets.

Problem Addressed

Many existing SLT systems rely on **sign gloss annotations**, which are intermediate labels representing each sign. Creating gloss annotations is labor-intensive and requires expert knowledge. Moreover, gloss-based approaches introduce an information bottleneck that limits the system's ability to understand sign language directly from videos.

Key Contribution

The paper proposes a Gloss-Free Sign Language Translation (GFSLT) framework that eliminates the need for gloss annotations. The model uses Visual-Language Pretraining (VLP) to learn relationships between sign language videos and textual sentences in a shared semantic space.

Methodology

- **Visual-Language Pretraining (VLP)**

The authors introduce a new pretraining strategy combining:

- **Contrastive language-image pretraining (CLIP)**
- **Masked self-supervised learning**

This allows the system to learn language-related visual features from sign language videos.

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- **GFSLT Architecture**

The model consists of:

- **Visual Encoder** – extracts features from sign language video frames
- **Transformer Encoder** – captures temporal relationships between gestures

- **Text Decoder** – generates the translated spoken language sentence

The pretrained visual encoder and text decoder are transferred to the final SLT model.

Datasets Used

The model is evaluated on two benchmark datasets:

- **RWTH-PHOENIX-Weather 2014T**
- **CSL-Daily Dataset**

These datasets contain sign language videos paired with spoken language sentences.

Results

The proposed approach significantly improves gloss-free SLT performance:

- **BLEU-4 score improvement:**
  - ≥ +5 on PHOENIX14T dataset
  - ≥ +3 on CSL-Daily dataset

The results show that visual-language pretraining effectively improves translation accuracy compared with previous gloss-free methods.

7. Conclusion

The study demonstrates that **visual-language pretraining combined with a gloss-free translation architecture** can significantly improve sign language translation performance. Removing gloss annotations also makes the system more scalable for real-world applications and reduces reliance on expert annotations.